

## Unit II Quantitative Research (12)

### 1.1 Meaning and characteristics of quantitative research

- Quantitative research is a method that uses numbers and statistics to test theories about customer/student/teacher/admin attitudes and behaviors.
- It helps researchers gather and analyze data systematically to gain valuable insights and draw evidence-based conclusions about customer preferences and trends.
- Researchers use online surveys, questionnaires, polls, and quizzes to question a large number of people to obtain measurable and bias-free data.
- In technical terms, quantitative research is mainly concerned with discovering facts about social phenomena while assuming a fixed and measurable reality.
- Let's consider an example.
- Suppose your qualitative analysis shows that your customers are looking for social media-based customer support. In that case, quantitative analysis will help you see how many of your customers are looking for this support.

## Quantitative research characteristics

Below are some of the characteristics of quantitative research.

### Large sample size

The ability to use larger sample sizes is undoubtedly one of the biggest perks of quantitative research.

### Measurability

Due to its quantitative nature, the data gathered through quantitative data collection methods is easily measurable.

### Close-ended questions

Quantitative research utilizes close-ended questions, which can be both beneficial and disadvantageous.

### Reusability

Since it doesn't involve open-ended questions, quantitative research results can be used in other similar research projects.

### Reliability

Quantitative data is considered more reliable since it is usually free of researcher bias.

### Generalization

Quantitative research uses larger sample sizes, so it is assumed that it can be generalized easily.

## 1.2 Random sampling: Simple and stratified

Sampling refers to the process of selecting a subset of individuals, items, or observations from a larger population to represent and make inferences about that population. In research and statistics, sampling is a crucial method used to gather data efficiently and effectively when it's not feasible or practical to study the entire population.

### Difference Between Simple and Stratified Quantitative Research

|      |                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i.   | <b>Methodology:</b>                                                                                                                                                                                              |
| a.   | <b>Simple Random Sample (SRS):</b> Involves randomly selecting individuals from the entire population without any prior categorization or grouping.                                                              |
| b.   | <b>Stratified Random Sample (SRS):</b> Involves dividing the population into smaller, homogeneous subgroups (strata) based on shared characteristics and then selecting samples from each stratum.               |
| ii.  | <b>Representation:</b>                                                                                                                                                                                           |
| a.   | <b>SRS:</b> Used to represent the entire data population without considering specific characteristics or differences.                                                                                            |
| b.   | <b>Stratified Random Sample:</b> Used to ensure representation of diverse characteristics within the population by sampling from each stratum.                                                                   |
| iii. | <b>Complexity and Cost:</b>                                                                                                                                                                                      |
| a.   | <b>SRS:</b> Simple and straightforward to implement, requiring fewer resources and less time.                                                                                                                    |
| b.   | <b>Stratified Random Sample:</b> More complex, time-consuming, and potentially expensive due to the need for stratification and separate sampling within each stratum.                                           |
| iv.  | <b>Suitability:</b>                                                                                                                                                                                              |
| a.   | <b>SRS:</b> Often used when there is limited information about the population, when the population is highly diverse and difficult to categorize, or when there is only one distinct characteristic of interest. |
| b.   | <b>Stratified Random Sample:</b> Preferred when the population exhibits significant variability across different characteristics or when specific subgroups need to be represented proportionally in the sample. |

Example: Suppose a marketing researcher wants to study the preferences for a new product among consumers in a city with diverse demographics.

- **Simple Random Sample (SRS): Example:** Suppose you want to conduct a survey to understand the preferences of students regarding extracurricular activities at a university. You assign a unique identifier (e.g., student ID number) to each student and then randomly select a certain number of students from the entire student population. Each

student has an equal probability of being selected, regardless of their year of study, major, or any other characteristic.

- **Stratified Random Sample: Description:** Stratified random sampling involves dividing the population into distinct subgroups or strata based on certain characteristics and then selecting samples from each stratum using simple random sampling. **Example:** Continuing with the university example, instead of selecting students randomly from the entire population, you decide to stratify the population based on the students' fields of study (e.g., engineering, humanities, sciences). After dividing the population into these strata, you then randomly select students from each stratum. For instance, if engineering students comprise 40% of the population, humanities students 30%, and sciences students 30%, you would ensure that your sample reflects these proportions by selecting a corresponding number of students from each field of study.

**Stratified Random Sample can be classified into two types:**

- **Proportionate Stratification:**
  - **Definition:** Proportionate stratification means allocating sample sizes to each subgroup (stratum) based on their proportion in the population.
  - **Example:** If a school has 60% boys and 40% girls, and you're conducting a survey on student preferences, a proportionate stratified sample would involve selecting 60 boys and 40 girls to ensure the sample reflects the gender proportions in the school.
- **Disproportionate Stratification:**
  - **Definition:** Disproportionate stratification involves intentionally allocating different sample sizes to subgroups (strata) that don't match their proportions in the population.
  - **Example:** If you're studying the academic performance of students across different grade levels in a school with 60% in Grade 9, 30% in Grade 10, and 10% in Grade 11, but you're particularly interested in Grade 11 students, you might disproportionately sample more Grade 11 students than their population proportion to gain deeper insights into that specific group. For instance, you might select 50% of Grade 11 students, 30% of Grade 10 students, and 20% of Grade 9 students, regardless of their population proportions.

### 1.3 Data collection tools: Questionnaire, test, scales and check list

These data collection tools are valuable for gathering quantitative data in a structured and systematic manner, allowing researchers to measure variables, test hypotheses, and analyze relationships within a research study. The choice of tool depends on the research objectives, the nature of the variables being measured, and the preferences of the researcher.

|      |                       |                                                                                                                                                                                                                                                                                                                                               |
|------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i.   | <b>Questionnaire:</b> |                                                                                                                                                                                                                                                                                                                                               |
|      | a.                    | <b>Description:</b> A questionnaire is a structured instrument used to collect data through written or electronic surveys consisting of a series of questions. Questions can be open-ended (allowing respondents to provide their own answers) or closed-ended (providing a set of response options to choose from).                          |
|      | b.                    | <b>Example:</b> In a study examining consumer preferences for a new product, a questionnaire may include questions about demographics, product features, purchasing behavior, and satisfaction levels. Respondents would rate their agreement with statements using a Likert scale or select from multiple-choice options.                    |
| ii.  | <b>Test:</b>          |                                                                                                                                                                                                                                                                                                                                               |
|      | a.                    | <b>Description:</b> A test is a standardized assessment tool used to measure an individual's knowledge, skills, abilities, or traits in a specific domain. Tests are often administered under controlled conditions and may consist of multiple-choice questions, short-answer questions, essays, or performance tasks.                       |
|      | b.                    | <b>Example:</b> In educational research, a standardized achievement test may be used to assess students' proficiency in mathematics. The test would contain a variety of mathematical problems covering different topics and difficulty levels, with scores indicating the students' level of mastery.                                        |
| iii. | <b>Scales:</b>        |                                                                                                                                                                                                                                                                                                                                               |
|      | a.                    | <b>Description:</b> Scales are instruments used to measure the intensity, frequency, or degree of a particular characteristic or construct. They typically consist of a series of items or statements to which respondents indicate their level of agreement or disagreement.                                                                 |
|      | b.                    | <b>Example:</b> In psychology, researchers may use a Likert scale to measure attitudes or perceptions. Participants would indicate their agreement with statements (e.g., "I feel confident in my abilities") on a scale ranging from "strongly disagree" to "strongly agree." The responses are then assigned numerical values for analysis. |
|      |                       | <b>Do you study at home ?</b>                                                                                                                                                                                                                                                                                                                 |
|      |                       | 1. Very Rarely 2 Rarely 3. Frequenty 5. Very Frequently 6 Daily                                                                                                                                                                                                                                                                               |
| iv.  | <b>Checklist:</b>     |                                                                                                                                                                                                                                                                                                                                               |
|      | a.                    | <b>Description:</b> A checklist is a simple tool used to record the presence or absence of specific items, behaviors, or criteria. It is often used to systematically collect observational data or to ensure that a set of tasks or requirements has been completed.                                                                         |
|      | b.                    | <b>Example:</b> In healthcare research, a checklist may be used to assess adherence to clinical guidelines during patient care. For example, a surgical safety checklist may include items such as "patient identity                                                                                                                          |

verified" and "site marked for surgery," with checkboxes for the surgical team to mark off as each step is completed.

#### 1.4 Statistical analysis: frequency, percentage, mean and standard deviation

##### i. **Frequency:**

- a. **Description:** Frequency refers to the count or number of times a particular value occurs in a dataset. It provides information about the distribution or occurrence of values within a variable.
- b. **Example:** In a survey dataset recording responses to a multiple-choice question asking participants about their favorite color, frequency analysis would show how many respondents selected each color option (e.g., red, blue, green, etc.).

##### ii. **Percentage:**

- a. **Description:** Percentage is a relative measure calculated as a proportion of a total expressed in terms of parts per hundred. It provides a way to express the frequency of occurrence of a value relative to the total sample size.
- b. **Example:** Continuing with the previous example, after conducting a frequency analysis of favorite colors, percentage analysis would express each color's frequency as a percentage of the total number of respondents. For instance, if 20 out of 100 respondents selected blue as their favorite color, the percentage of respondents who chose blue would be 20%.

##### iii. **Mean:**

- a. **Description:** The mean, also known as the average, is a measure of central tendency calculated by summing all values in a dataset and dividing by the total number of values. It represents the typical or average value of the dataset.
- b. **Example:** Consider a dataset containing the ages of five individuals: 25, 30, 35, 40, and 45. The mean age would be calculated as  $(25 + 30 + 35 + 40 + 45) / 5 = 35$ . Thus, the mean age of the individuals in the dataset is 35 years.

##### iv. **Standard Deviation:**

- a. **Description:** The standard deviation measures the dispersion or spread of values around the mean in a dataset. It provides information about the variability or consistency of values within the dataset.

**Example:** Using the same dataset of ages mentioned earlier, the standard deviation would quantify how much each individual's age deviates from the mean age of 35. A higher standard deviation indicates greater variability among the ages, while a lower standard deviation indicates more consistency.

#### 1.5 Types of quantitative research

|      |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i.   | <b>Experimental Research:</b>                                                                                                                                                                                                                                                                                                                                                                                       |
| a.   | <b>Description:</b> Experimental research involves manipulating one or more variables to observe the effect on another variable, while controlling for extraneous factors. It aims to establish cause-and-effect relationships between variables.                                                                                                                                                                   |
| b.   | <b>Example:</b> A pharmaceutical company conducts an experimental study to test the effectiveness of a new drug in reducing blood pressure. Participants are randomly assigned to two groups: one group receives the new drug (experimental group), while the other group receives a placebo (control group). After a specified period, changes in blood pressure are measured and compared between the two groups. |
| ii.  | <b>Survey Research:</b>                                                                                                                                                                                                                                                                                                                                                                                             |
| a.   | <b>Description:</b> Survey research involves collecting data from a sample of individuals using standardized questionnaires or interviews. It aims to gather information about opinions, attitudes, behaviors, or characteristics of a population.                                                                                                                                                                  |
| b.   | <b>Example:</b> A marketing firm conducts a survey to assess customer satisfaction with a new product. They distribute a questionnaire to a random sample of customers, asking about their overall satisfaction, likelihood of repurchase, and feedback on specific product features. The responses are then analyzed to identify patterns and trends in customer satisfaction.                                     |
| iii. | <b>Correlational Research:</b>                                                                                                                                                                                                                                                                                                                                                                                      |
| a.   | <b>Description:</b> Correlational research examines the relationship between two or more variables without manipulating them. It aims to determine whether and to what extent changes in one variable are associated with changes in another variable.                                                                                                                                                              |
| b.   | <b>Example:</b> A researcher conducts a correlational study to investigate the relationship between study hours and academic performance among college students. They collect data on the number of hours students study each week and their GPA scores. By analyzing the data, the researcher can determine whether there is a positive or negative correlation between study hours and academic performance.      |